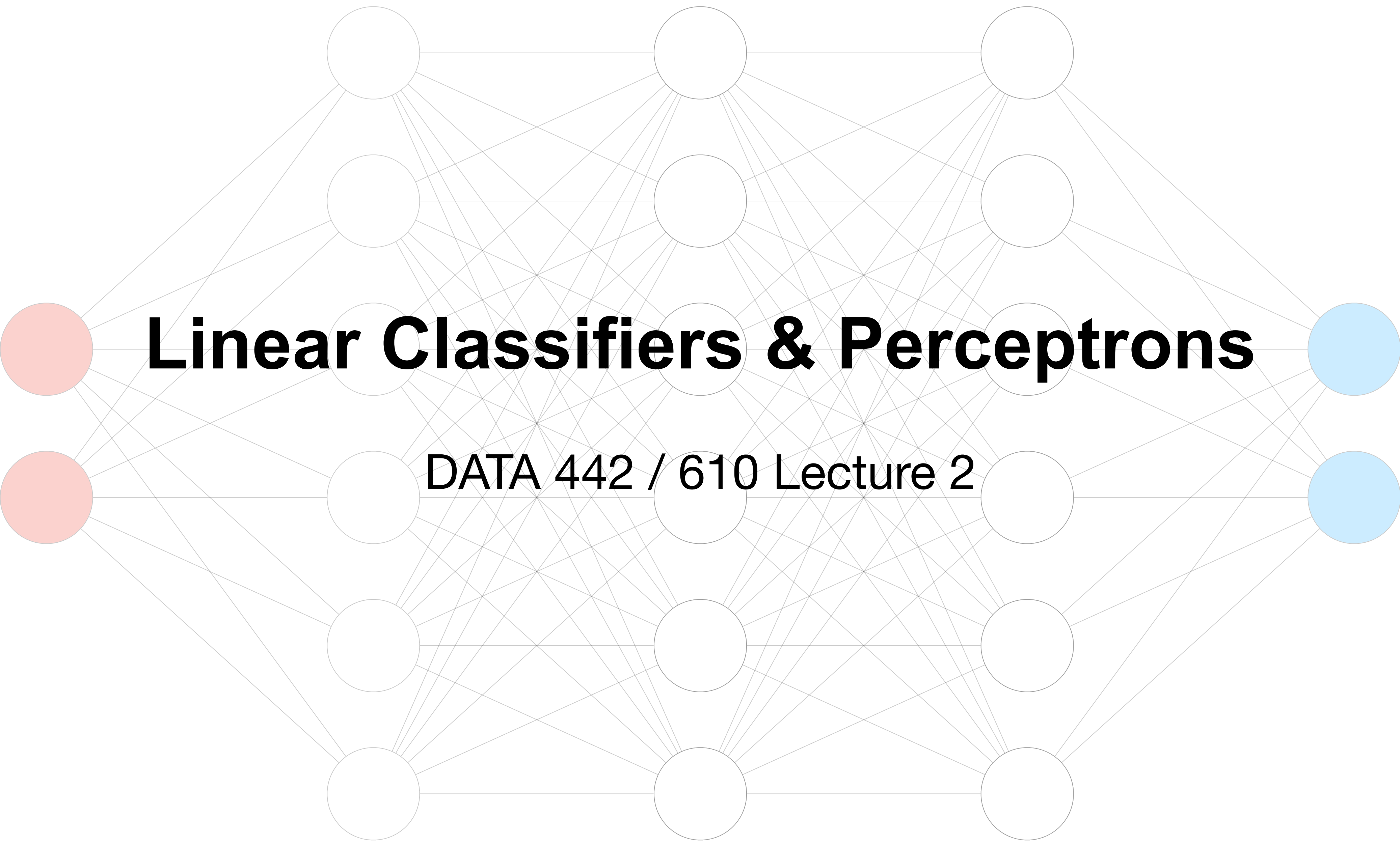




Linear Classifiers & Perceptrons

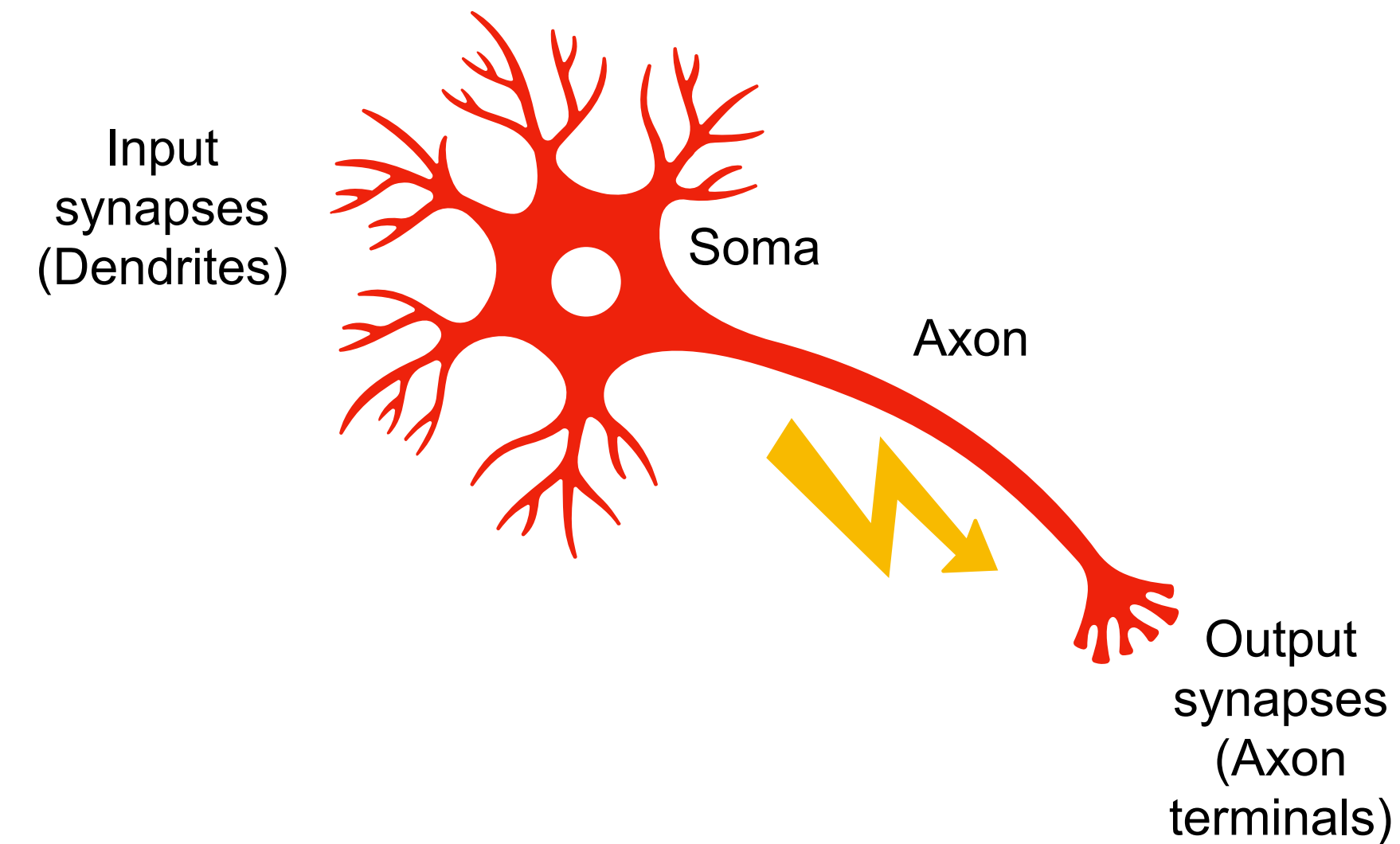
DATA 442 / 610 Lecture 2

Outline

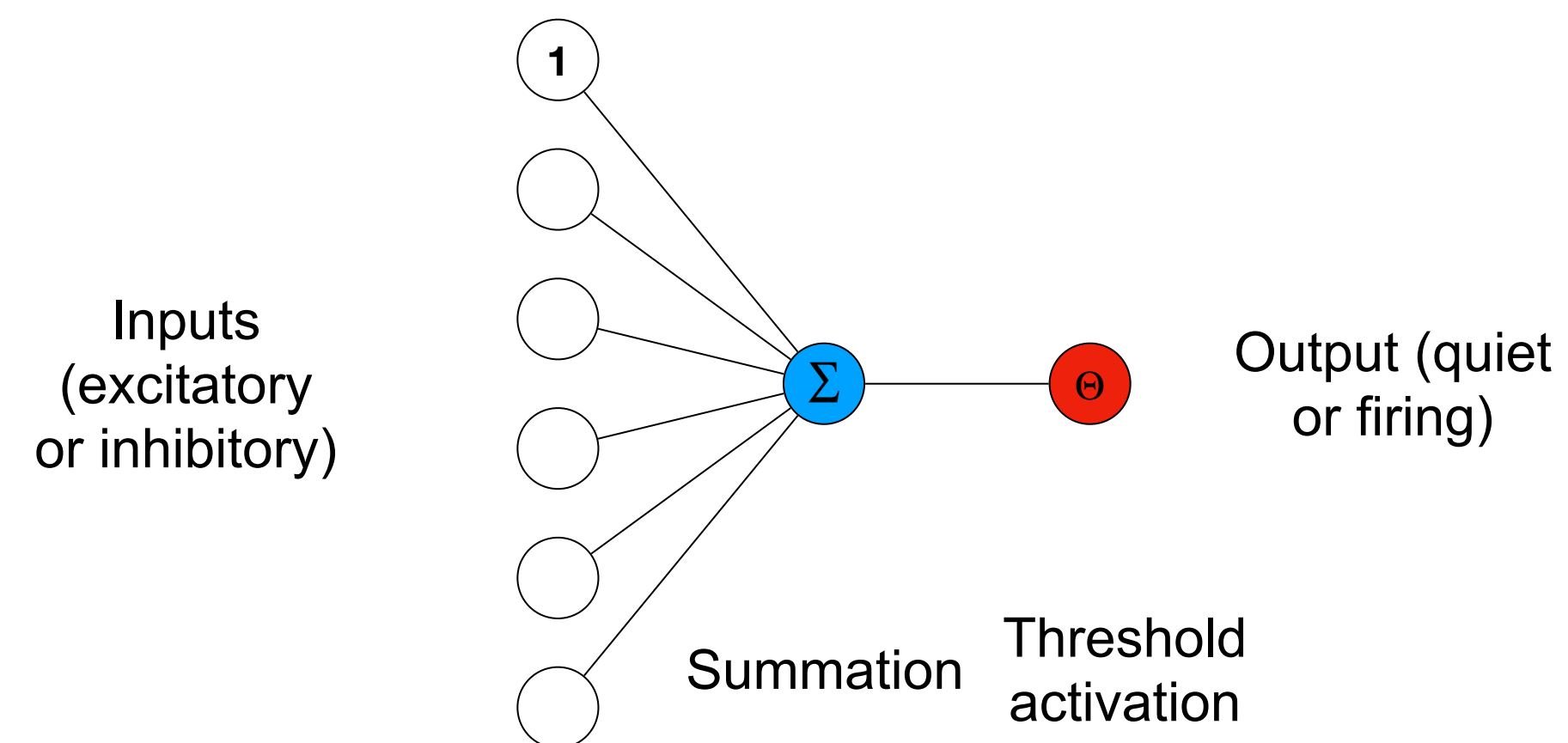
- Linear Classification
 - Perceptron models
 - Proof of Convergence
 - Code demo
 - Adaline

Artificial neurons and the perceptron model

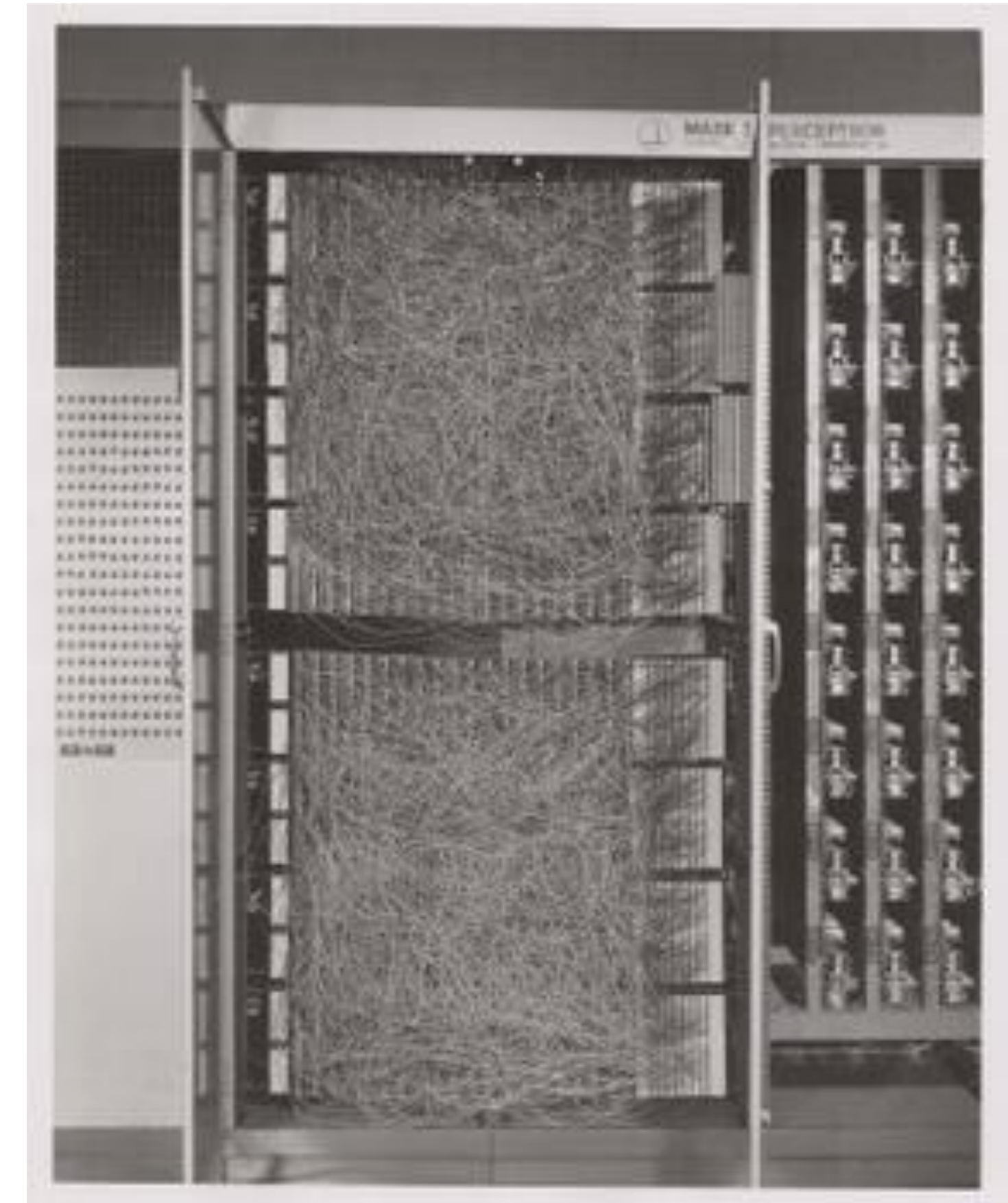
Biological Neuron



Artificial Neuron (McCulloch/Pitts, 1943)

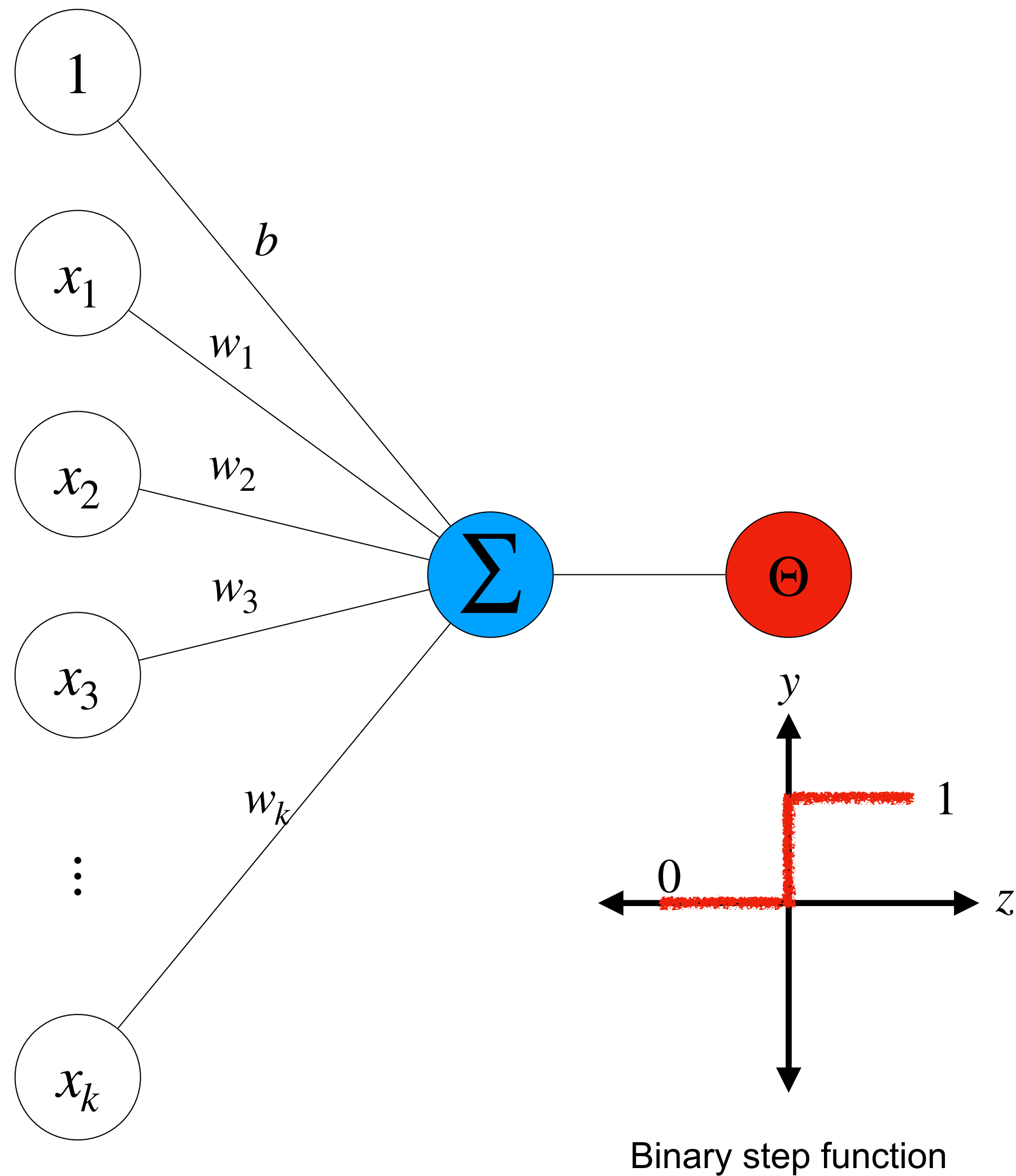


Perceptron (Rosenblatt, 1957-58)



Mark I Perceptron machine (first hardware implementation of learning with an artificial neural network)

Perceptrons for binary classification



Suppose we have k -dimensional features

$$\text{Let } z = w_1 x_1 + w_2 x_2 + \dots + w_k x_k + b = \mathbf{w}^T \mathbf{x} + b$$

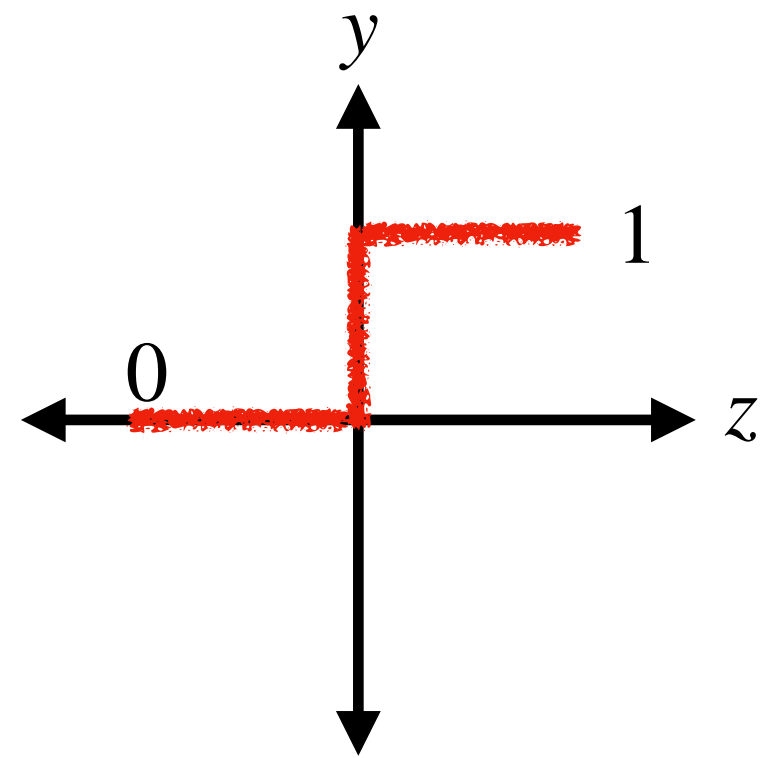
z is a **preactivation** to which we apply an **activation function**

$$\sigma(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We define our predicted class label $\hat{y} = \sigma(z)$ for each example

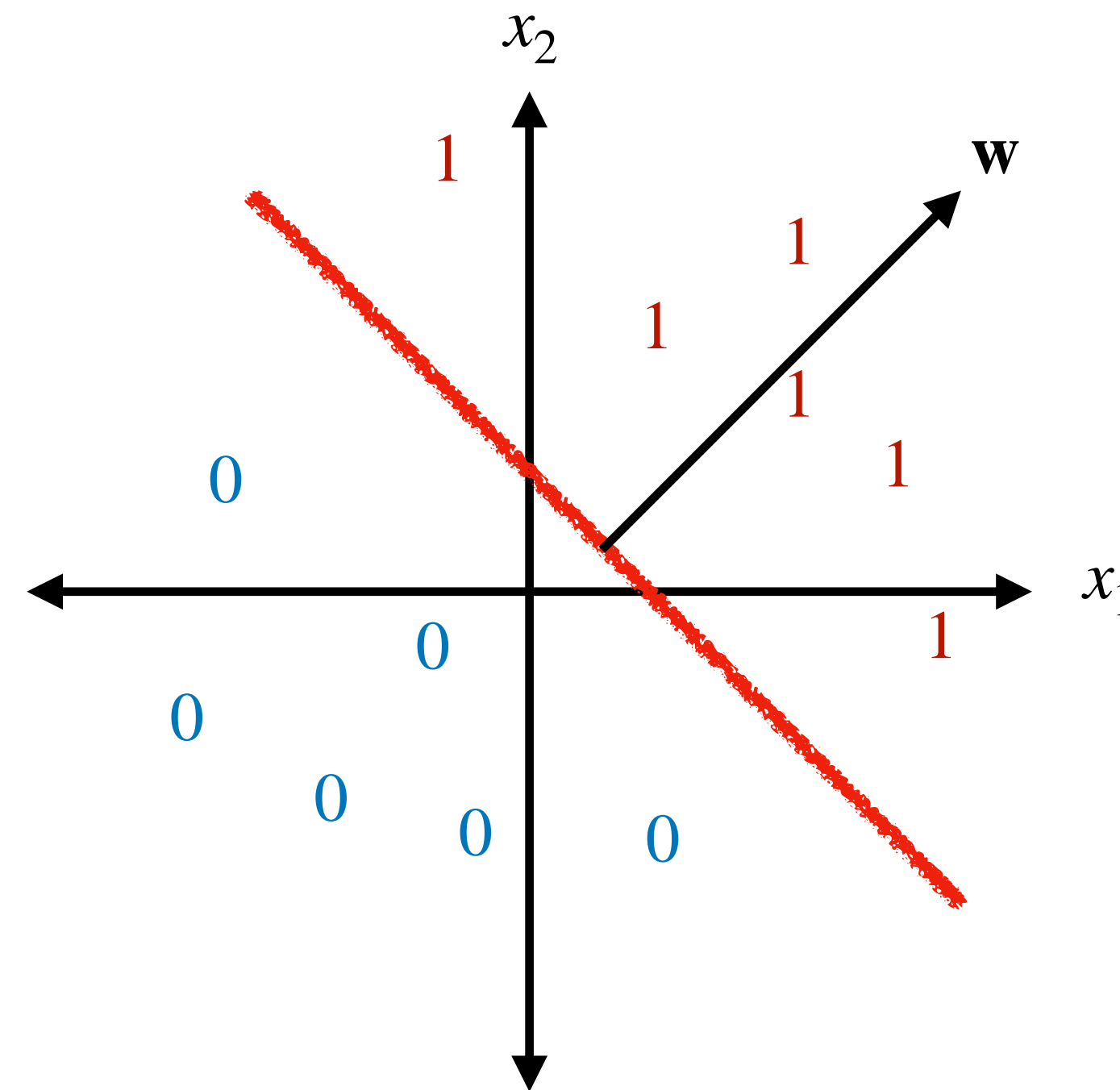
Our goal is to find the weight vector $\mathbf{w} \in \mathbb{R}^k$ that minimizes the errors between the predicted class $\hat{y}^i$ and true label y^i for each example $0 \leq i < N$

Geometric interpretation of a perceptron



The linear perceptron with binary step function learns a linear **decision boundary** between two classes

The decision boundary is a **hyperplane** in the k -dimensional feature space



The weight vector $\mathbf{w}$ is *perpendicular* to the decision boundary

Why?

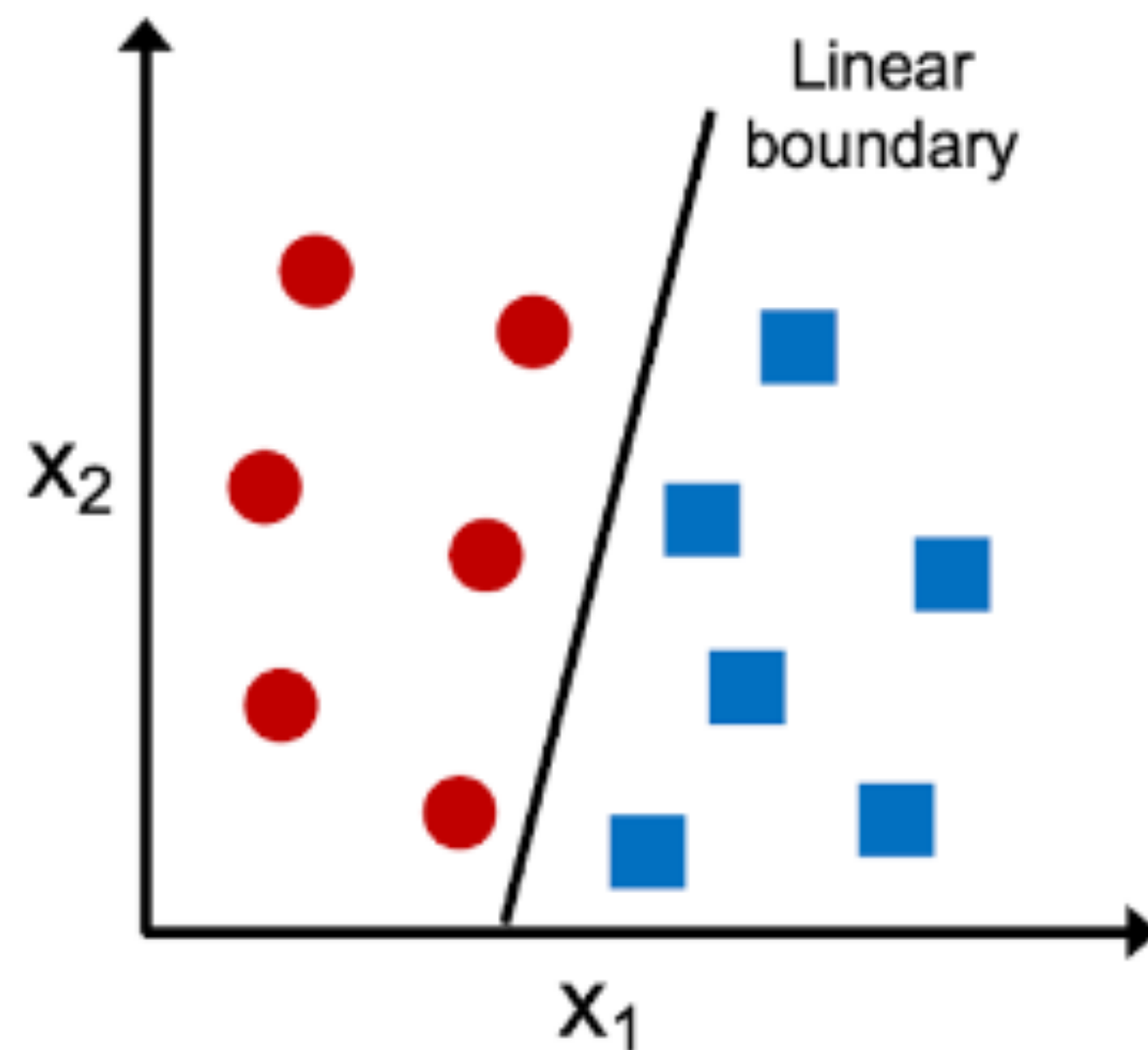
A plane is defined using a point $\mathbf{x}_0$ and a normal vector $\mathbf{n}$ as the set of points $\mathbf{x}$ such that $\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0$

A perceptron learns $\mathbf{w}$, b corresponding to the normal $\mathbf{n}$ and offset $\mathbf{n} \cdot \mathbf{x}_0$ for the decision boundary

Perceptrons are for linearly-separable classes

Linearly separable

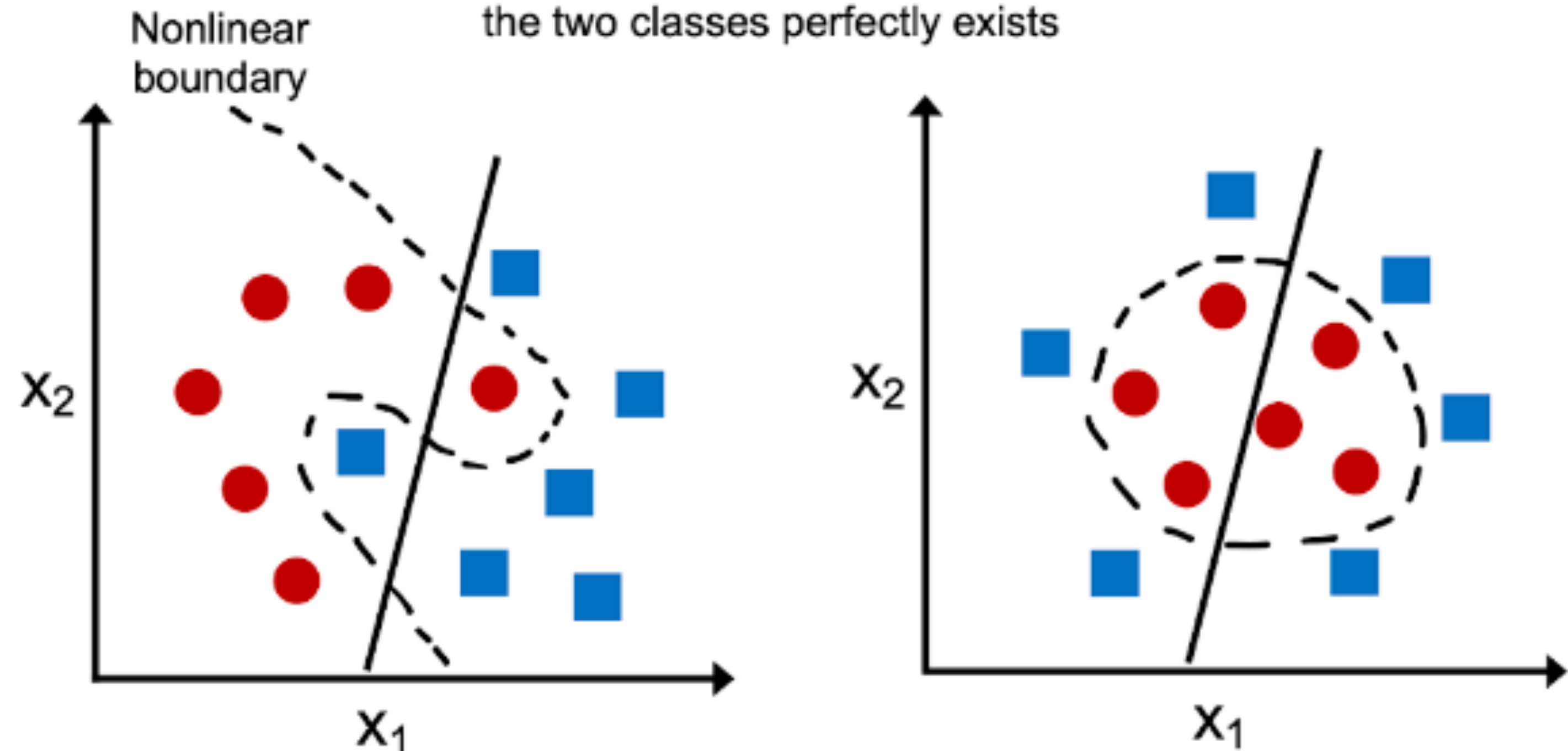
A linear decision boundary that separates the two classes exists



Perceptrons work for these

Not linearly separable

No linear decision boundary that separates the two classes perfectly exists



Perceptrons are only guaranteed to converge for linearly separable classes!

Courtesy of C. Fanelli

Perceptron Learning Rule

Goal: Find $\mathbf{w}$ to minimize the number of perceptron errors $y \neq \hat{y}$

Algorithm: For each training example $\mathbf{x}^i$

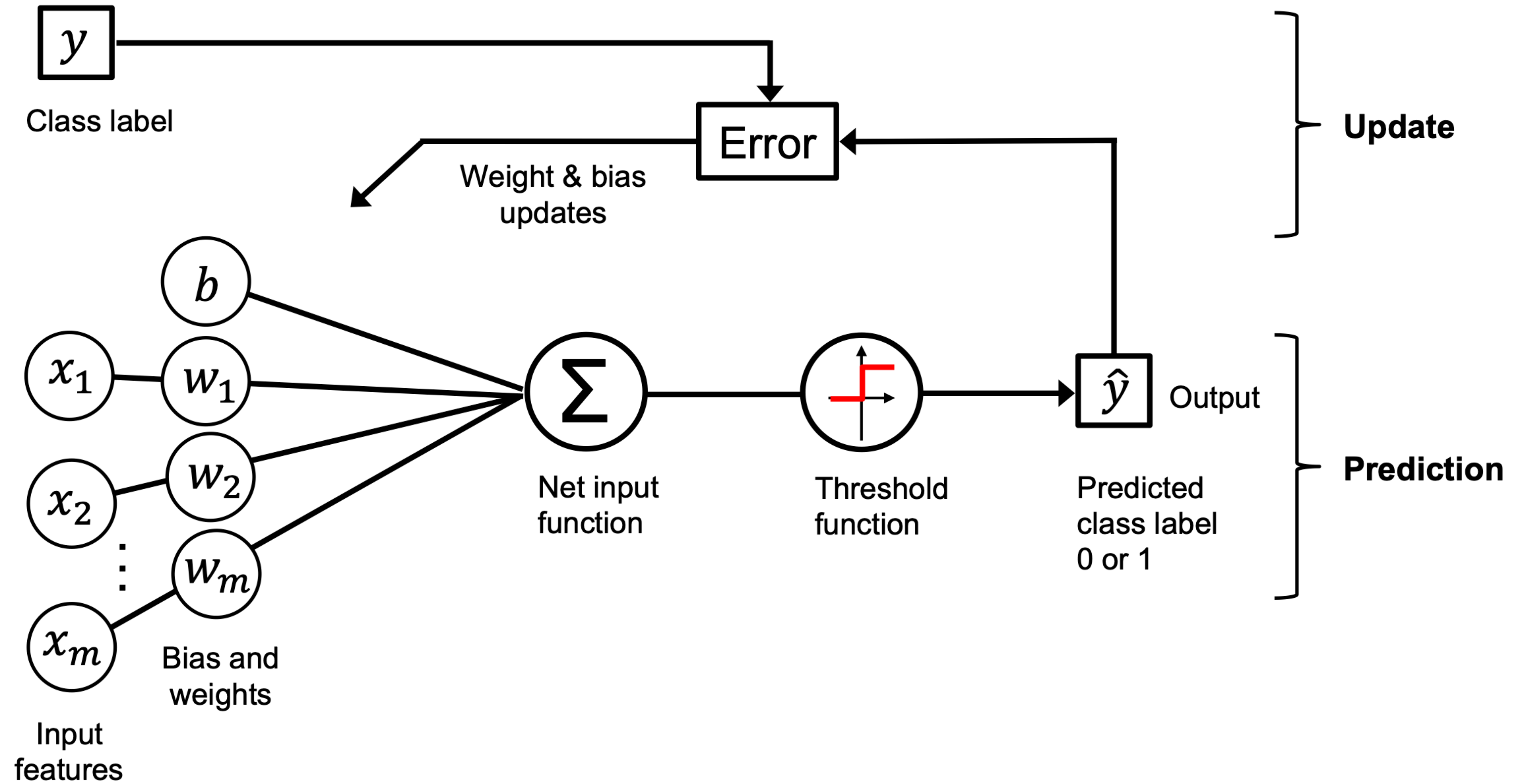
1. Compute the output $\hat{y}^i = \sigma(\mathbf{w}^\top \mathbf{x}^i + b)$
2. Update the weights and bias

$$w_j \leftarrow w_j + \Delta w_j$$

$$b \leftarrow b + \Delta b$$

Proof of Convergence

Perceptron Flowchart



Hands-On Demo: Binary Classification with Perceptron



Perceptron: Takeaways

The good

- Performs binary classification
- Guaranteed convergence for linearly separable classes

The bad

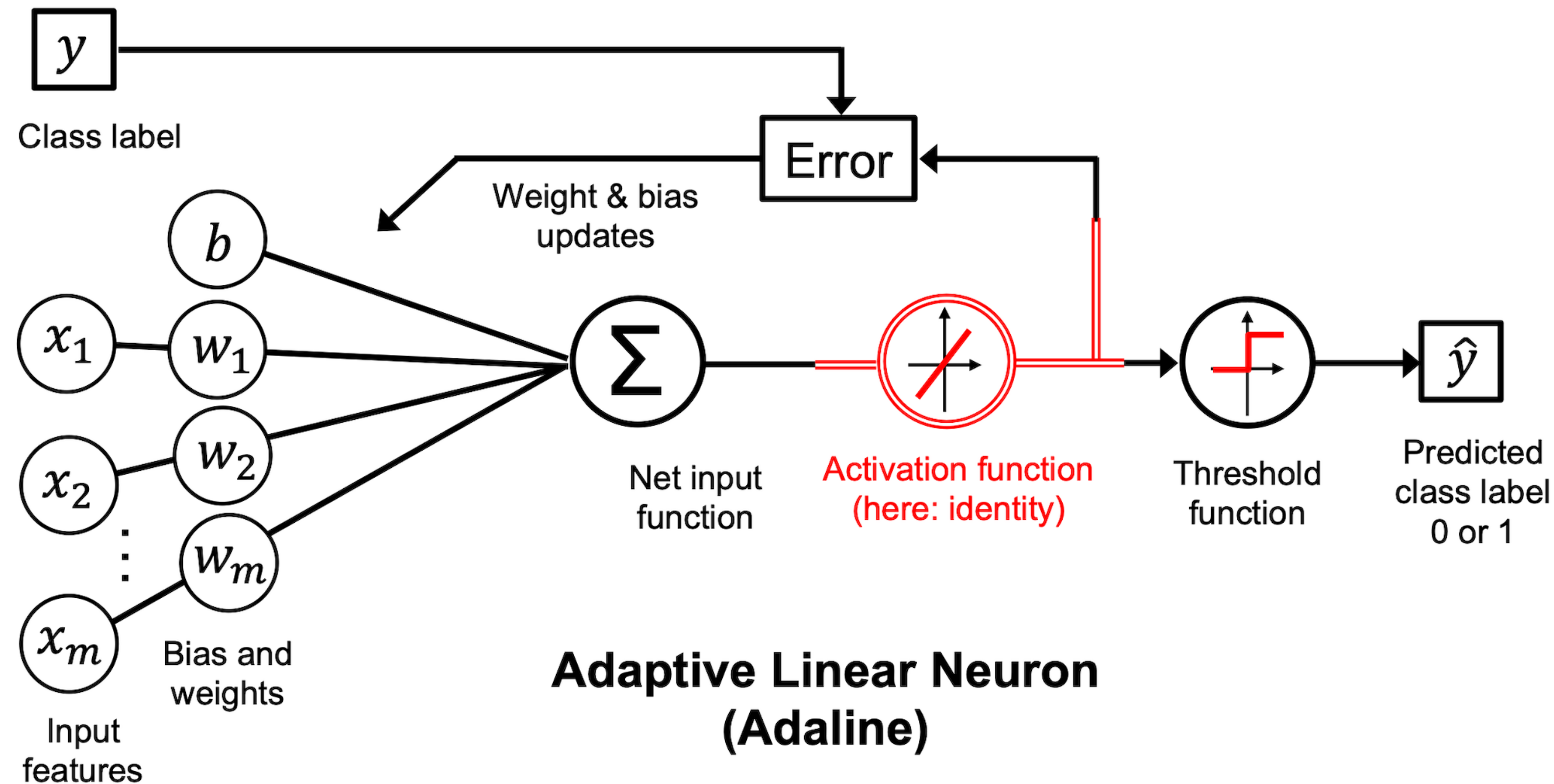
- Does not allow nonlinear decision boundaries

The ugly

- Hand-wavy derivation of weight update

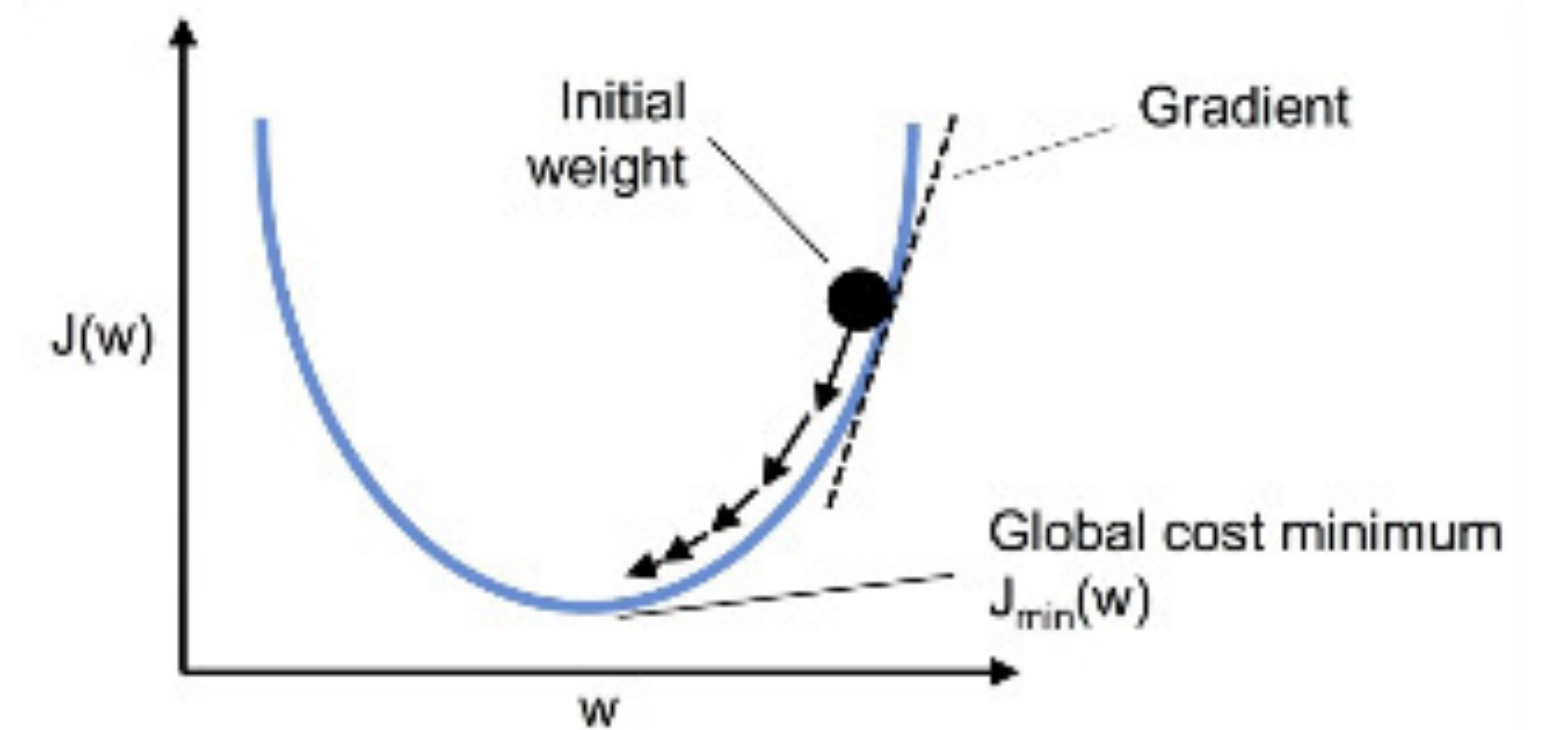
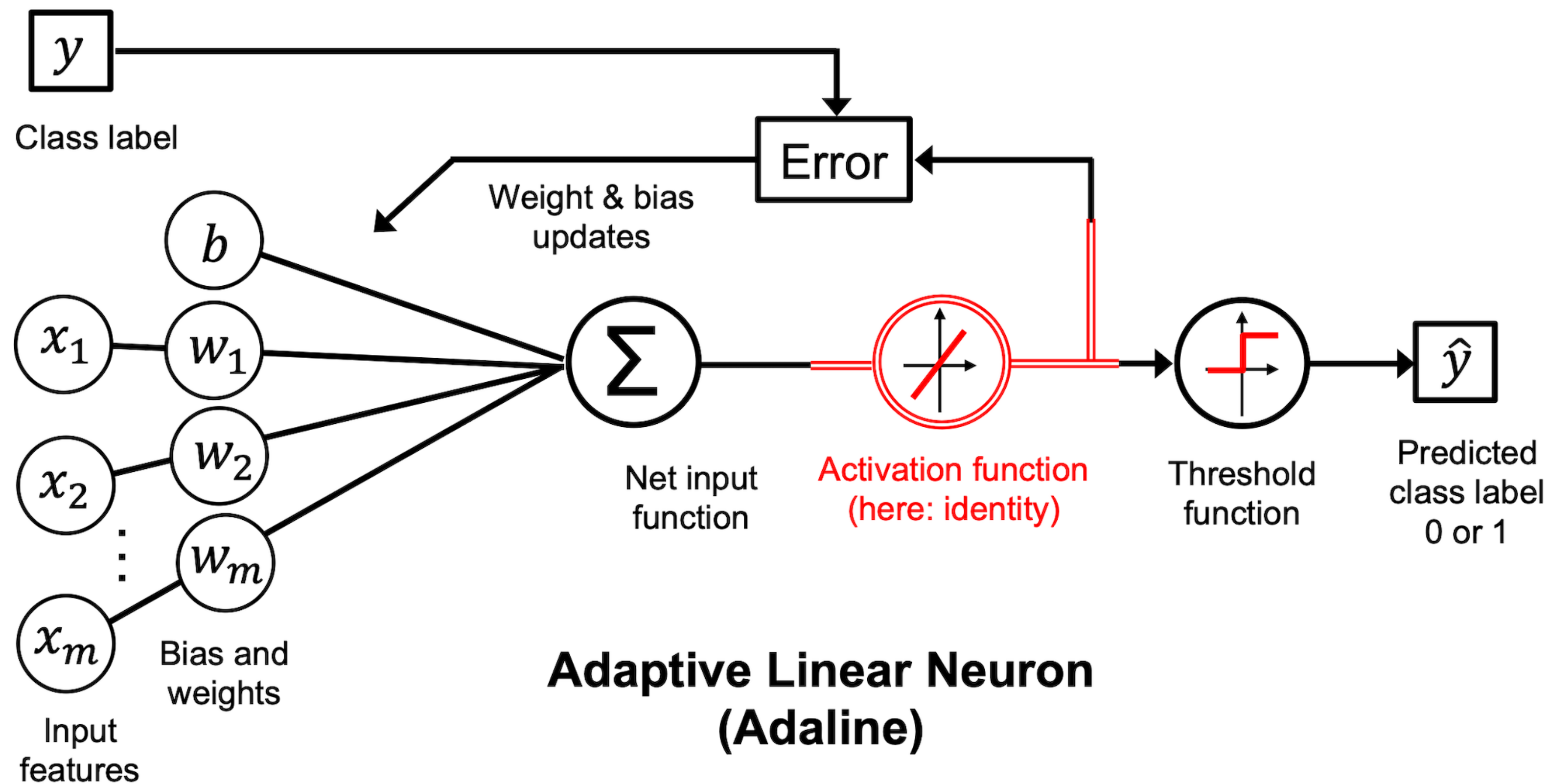
ADALINE (ADaptive Linear NEurons) - 1960

- Adaline is an improvement over the perceptron that minimizes a **continuous error** function rather than a binary classification output
- Adaline trains using a **linear activation** $\sigma(z) = z$ and predicts using a threshold function



Courtesy of C. Fanelli

Adaline Flowchart



Mean Squared Error (MSE) Loss

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y^i - \sigma(z^i))^2$$

Courtesy of C. Fanelli

Gradient Descent with Adaline

Goal: Find $\mathbf{w}$ to minimize the Adaline loss $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y^i - \sigma(z^i))^2$

Algorithm: For each training example $\mathbf{x}^i$

1. Compute the output $\sigma(z^i) = \mathbf{w}^\top \mathbf{x}^i + b$
2. Update the weights and bias

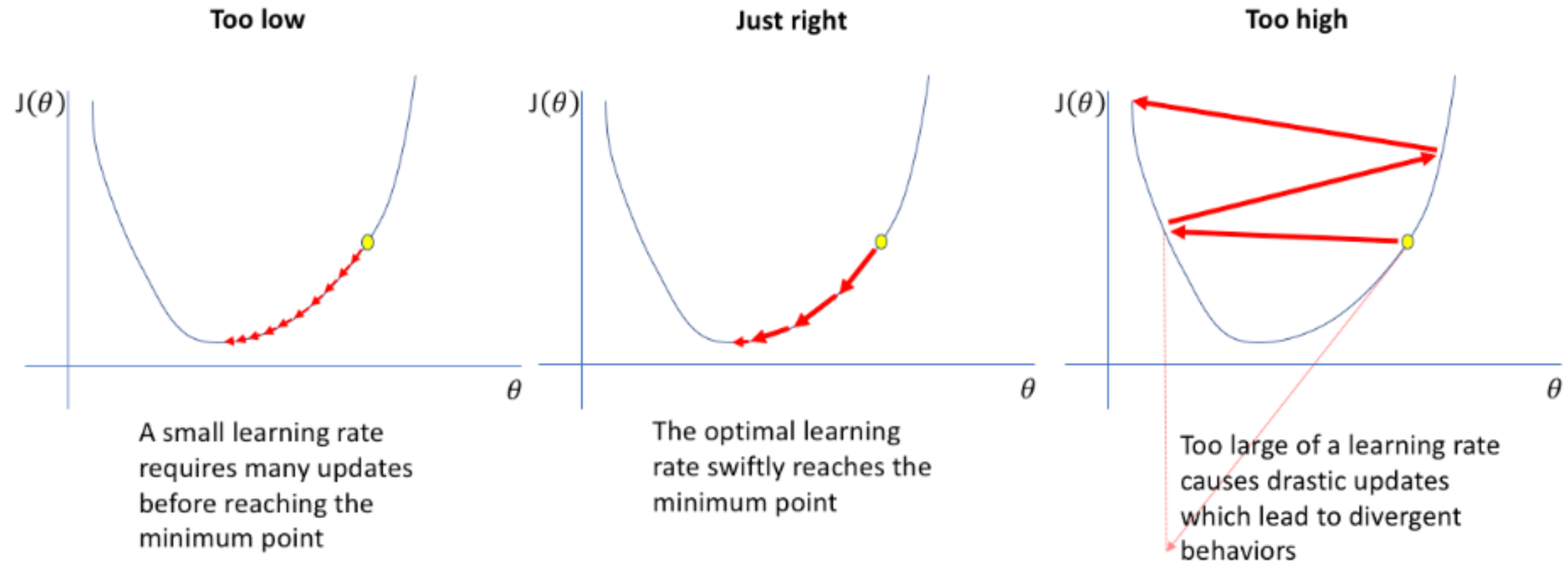
$$w_j \leftarrow w_j + \Delta w_j$$

$$b \leftarrow b + \Delta b$$

Hands-on Demo: Binary Classification with Adaline



The importance of learning rate



We will return to this point again, and again, and again

Courtesy of C. Fanelli

Adaline: Takeaways

The good

- Update rule comes from “somewhere”
- Continuous error function allows gradient descent

The bad

- See “Perceptron”

The ugly

- Learning rate can make or break fitting

That's all!

- Slides will be posted on the course website: <https://colen-teaching.github.io/DATA442-Fall-2026>
- Lab demos will also be posted
- Be on the lookout for PS 0: A **Bonus** Assignment